

MA 525: Heuristic Optimization

Course Code	MA 525
Course Name	Heuristic Optimization
L-T-P-C	3-0-0-3
Prerequisites	IC150, IC111 or equivalent / instructor's consent.
Intended for	B.Tech. 3rd, 4th year, M.S./J.M.Tech./M.Sc., Ph.D.
Distribution	Elective
Approval	20th BoA (Dec 2017); 15th Senate (Feb 2018)

Course Contents

- **Introduction** [2 Lectures]
Introduction to optimization, Local and Global Minima, Classical Optimization Techniques, Heuristic Optimization techniques.
- **Random number generations** [7 Lectures]
Random numbers of a given distribution, properties and statistical tests. Simulation of random number generators.
- **Benchmarks and algorithms comparisons** [4 Lectures]
Parameter settings and statistical criterion for comparison of various algorithms, parametric and non parametric tests, non statistical measures and issues with them.
- **Continuous Optimization** [8 Lectures]
Evolutionary Techniques, Swarm based Techniques and other nature inspired techniques. Theoretical foundations of various techniques. Implementation issues with various techniques and their comparisons based on benchmarks
- **Discrete and combinatorial Optimization** [4 Lectures]
Heuristic optimization approaches for discrete, mixed continuous discrete and combinatorial problems. Application to solve Knapsack, TSP, Network Flow problems, Submodular Functions under Matroid Constraints.
- **Estimation of Distribution Algorithms** [4 Lectures]
EDA for discrete optimization and their comparisons, continuous EDA, Application of discrete and continuous EDA in optimization and machine learning.
- **Hybrid techniques** [4 Lectures]
Local search methods and their advantages. Hybrid optimization techniques. Use of hybrid techniques and their application.
- **Constraint handling techniques** [6 Lectures]
Problems with inequality and equality constraints. Methods based on rejection strategies, repair strategies, specialized operators. Penalty parameter based and penalty parameter less approaches. Approaches for handling equality constraints. Implementation of various constraint handling techniques and their comparison over various practical and benchmark problems.
- **Multi objective optimization** [6 Lectures]
Various approaches to handle multiple objectives, Pareto Optimality. Dominance and decomposition based approaches. Hybrid techniques. Bi-level optimization. Theoretical Foundations and Applications to engineering and finance.

Text Books

1. Engelbrecht, Andries P., **Fundamentals of computational swarm intelligence**, John Wiley & Sons, 2006.
2. Deb, K., **Multi-objective optimization using evolutionary algorithms**, John Wiley & Sons, 2001.

References

1. Mezura-Montes, E. (Ed.), **Constraint-Handling in Evolutionary Optimization** Constraint Handling in Evolutionary Optimization, Studies in Computational Intelligence, vol. 198, Springer-Verlag, 2009.
2. Eiben, A.E. and Smith, J. E., **Introduction to Evolutionary Computing**, Springer, 2003.
3. Niederreiter, H. **Random number generation and quasi-Monte Carlo methods**, Society for Industrial and Applied Mathematics, 1992.
4. Coello, C. A. C., Lamont, G. S., and Veldhuizen, D. A. V., **Evolutionary algorithms for solving multi-objective problems**, Vol. 5, Springer, 2007.
5. Datta, R., and Deb, K. (Eds.), **Evolutionary constrained optimization**, Springer, 2014.
6. Lobo, F. J., Lima, C. F., and Michalewicz, Z. (Eds.), **Parameter setting in evolutionary algorithms**, Vol. 54, Springer Science & Business Media, 2007.
7. Blum, C., Roli, A. and Sampels, M. (Eds.), **Hybrid metaheuristics: an emerging approach to optimization**, Springer, 2008.
8. Larranaga, P. and Lozano, J. A. (Eds.). **Estimation of distribution algorithms: A new tool for evolutionary computation**, Springer Science & Business Media, 2012.
9. Clerc, M., **Guided randomness in optimization**, Vol. 1, John Wiley & Sons, 2015.
10. Wolsey, L.A., and Nemhauser, G. L., **Integer and Combinatorial Optimization**, Wiley, 1999.
11. Zbigniew M. and Fogel, D., **How to Solve it: Modern Heuristics**, Springer Verlag, 2000.